

The Contaminated City

Chief Scientific Officer • Riverton Contamination Response

10 DAYS • 41 STOPS • GRADE 6 • CONTAMCITY_MS

THE OPENING CARD

A freight yard beside the river burned last night, and nobody can say what came off it. You are the city's chief scientist, which means you decide what gets measured, and what the city gets told. Two neighbourhoods sit downwind of a cloud nobody has named. The drinking water for a hundred and forty thousand people is drawn from the river just below the yard, and the utility has shut the intake. So the city is drinking what it has stored, and what it has stored is measured in days.

The Unknown Containers

BEFORE THE DAY — GREET

Six offices in one city, and nobody has met

The city laboratory, the water company and the health office all take part of this job. By Friday one of them has to sign for another's number. Put a name to as many of them as you can now, because a signature you cannot picture is one you check twice.

PLAN CARD 71W

Day one. The freight yard fire has been out for six hours. A row of damaged containers sits eighty metres from the river, and nobody knows what was in them. Three names are going around, and they do not agree. Adaeze Okonjo, who runs the laboratory, wants two tests before anybody writes a name down. The fire chief wants a name now. Today you build a short list you would defend.

BRIEFING 41W

A freight-yard fire damages several unlabeled containers beside Riverton's river. The response team must establish chemical identity and hazard classes before firefighters, residents, and the water utility act on rumors. A wrong identity can cause an incompatible firefighting or treatment decision.

WHAT YOU DECIDE 11W

Create a provisional identity list and choose the next discriminating measurements.

1.1 Read the label, not the rumour

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM ·
PROTOCOL

SCENE 36W

A fire at the freight yard has damaged several unlabelled drums, eighty metres from the river. Three different guesses are already on the radio. Firefighters are choosing what to spray based on whatever you tell them.

GUIDE 53W

The left side is four kinds of record. The right side is four things you could say from one. Ask of each record what it really pins down. Which elements? How many of each? What charge? A trade name pins down none of them. Firefighters are choosing what to spray from your answer.

ASK 12W

Match each thing on the record to what it lets you conclude.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a chemical formula says which elements are in something, and how many of each · dissolving, and why some things dissolve and others do not — taken as read

VERDICT 78W

A formula says which elements are in something and how many of each. That is what decides how it behaves: what it dissolves in, what it reacts with, what it does to water. Change one of those numbers and it is a different substance with different dangers, even though the elements are the same. A trade name carries none of that. It is a word somebody chose, and some of the substances on this list react with water.

TAKEAWAY 20W

A name is a label. A formula says what something is made of, which is what decides how it behaves.

1.2 From damaged drum to a first answer

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · SEQUENCE

SCENE 46W

In the mobile laboratory, the first sealed drum sample is on the bench. The markings on the outside are readable now, and opening the drum will scuff them off. Opening it also lets the air above the liquid escape. There is enough liquid for two tests.

GUIDE 50W

Every one of these can be done right now. So this is not about time. It is about what you can still do afterwards. Ask what each step uses up. The markings? The air above the liquid? The drops? The sample itself? There is only enough liquid for two tests.

ASK 13W

Put the four steps in order of what each one costs, cheapest first.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

some tests leave a sample as they found it, and some use it up

VERDICT 73W

The order is set by what each step costs. The markings on the outside survive only until somebody opens the drum, so they get photographed first, and it costs nothing. Sampling the air above the liquid takes what has already escaped, without opening anything. A test that gives the sample back can be repeated all week if it goes wrong. The test that uses the sample up answers best, and it answers once.

TAKEAWAY 14W

Do the tests that cost nothing before the ones that cannot be taken back.

1.3 Float or sink

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · CHOICE

SCENE 33W

One drum is leaking into the bund. Nobody knows what the liquid is yet. Adaeze Okonjo needs to know whether it will float on the water already lying there, or sink under it.

GUIDE 52W

All four options are things you could measure about this liquid. Only one of them decides floating. Floating is about how heavy a litre is, next to a litre of water. Colour and smell tell you other things. So does the amount in the drum. The bund is filling while you decide.

ASK 9W

Which measurement decides whether the liquid floats on water?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a litre of water weighs about a kilogram

VERDICT 101W

Floating is a comparison. A litre of water weighs about a kilogram, so a litre of anything lighter than that rides on top and a litre of anything heavier goes under. Nothing else on the list decides it. Colour and smell say nothing about weight. The amount in the drum does not change it either — a full drum and a spoonful of the same liquid float or sink the same way. The answer matters tonight, because a floating layer spreads across the bund and reaches the drain, while a sinking one sits at the bottom where the pump inlet is.

TAKEAWAY 20W

Whether something floats is a comparison, not a property: it is heavier or lighter than the same amount of water.

Create a provisional identity list and choose the next discriminating measurements.

The Vapor Cloud

BEFORE THE DAY — TRIAL

Walk the river bank before you take a sample

The city sits along the river, and every sample means something different depending on where the water was standing. Walk the near bank once. A map does not show you which outfall is above which bridge.

PLAN CARD 74W

Day two, first light. The wind has turned, and a colourless cloud is drifting off the yard toward two neighbourhoods. Miklos Varga's forecast and the fire chief's map disagree about where it goes by eleven. Sipho Mbeki has to tell residents something by eight. Today you work out how much air that release can fill, and what the warming morning does to it. Warn too small an area and somebody is standing in it.

BRIEFING 44W

A colorless plume moves toward two neighborhoods as the weather changes. The team must estimate expansion, density, and direction without treating an ideal-gas model as a complete atmospheric forecast. A delayed evacuation risks exposure; an overbroad evacuation can block emergency access and overwhelm shelters.

WHAT YOU DECIDE 11W

Provide a bounded plume assessment and decision triggers for evacuation zones.

2.1 How long before it reaches the school

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · BALLPARK

SCENE 31W

A colourless cloud is drifting away from the yard toward two neighbourhoods. Police are holding the evacuation roads open. The wind is steady, and there is a school two kilometres downwind.

GUIDE 46W

Four numbers, and two of them are distances to other places. One is how wide the cloud is, which does not change when it arrives. Ask of each number what it measures. A distance? A speed? A size? Police are holding roads open on this answer.

ASK 8W

How long before the cloud reaches the school?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a cloud drifts at roughly the speed of the wind carrying it

VERDICT 76W

A cloud is carried by the wind, so it travels at roughly the wind's speed. Two thousand metres at four metres a second is five hundred seconds, which is a little over eight minutes. That is the whole decision. Eight minutes is enough to move a school if somebody starts now, and not enough if the first thing anybody does is argue. And if the wind picks up, the eight minutes shortens without anybody being told.

TAKEAWAY 16W

A hazard has a speed, and a speed turns a distance into the time you have.

2.2 What changes the cloud

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · PROTOCOL

SCENE 25W

The weather station shows the temperature rising and the pressure falling, with the wind picking up. Crews report vapour pooling in a low channel downwind.

GUIDE 50W

Four changes in the field. Two of them change how much room the gas takes up. Two of them change where the gas goes. Ask which of the two each change does. Get that wrong and the map moves for the wrong reason. Vapour is pooling in a low channel.

ASK 10W

Match each change in the field to what it does.

BEHIND THE BUTTON 161W

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How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a gas has a state, and it also has a place, and those change for different reasons

VERDICT 78W

Two different things are happening at once. Warming a gas makes it take up more room, and dropping the pressure around it does the same. That is about the gas itself. Whether it hugs the ground or lifts away depends on how heavy it is compared with air, which is a different question again. And where it goes is the wind's business entirely. Nothing about the gas changed when the wind turned. The place it was heading did.

TAKEAWAY 20W

Some things change what the gas is doing. Others change where it is going. They are not the same question.

2.3 Why the drum is worse in the afternoon

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · CHOICE

SCENE 30W

The same open drum reads three times the vapour at two in the afternoon as it did at six in the morning. Nothing has been added and nothing taken out.

GUIDE 50W

All four options would explain a bigger reading. They differ in what changed. One changes the liquid. One changes the instrument. One changes how much is left. One changes only how much sits in the air. Nothing was added and nothing taken out. That is the fact that decides it.

ASK 9W

Same drum, same liquid, three times the vapour. Why?

BEHIND THE BUTTON 169W

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Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a liquid is always sending some of itself into the air above it

VERDICT 78W

A liquid is always losing molecules into the air above it, and getting some back. How many are up there at once depends on temperature, and it climbs steeply. Ten or fifteen degrees can double it. So the drum that was safe to stand near at dawn is not the same drum at two in the afternoon, with nothing having been added or taken away. It is why measurements get taken at the hottest part of the day.

TAKEAWAY 13W

A measurement of a hazard belongs to the conditions it was taken in.

2.4 Water helps, or water makes it worse

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · BELT

SCENE 37W

The contractor lifts at four and the yard has to be sorted first. Some of these drums are made safer by water. Others react with water, and a hose is the worst thing that could reach them.

GUIDE 34W

Two bins. Ask what happens when water reaches it. Something that dissolves is made weaker by water. Something that reacts with water makes heat or gas, and then the hose has made things worse.

ASK 12W

Send each drum to the bin that says what water would do.

BEHIND THE BUTTON 66W

Why this is the first question. Water is what a fire crew has, so water is what usually arrives. For most of a yard that is right, and for a short list it is the worst possible move.

What reacting looks like. Heat, sometimes enough to boil the water. Gas coming off that was not there before. Both happen in seconds and neither can be undone.

TAKES AS READ

some things dissolve in water and some react with it

VERDICT 153W

Water is the usual answer to a spill because it is what everybody has. For most of a yard it is the right one. A salt or a weak acid gets weaker as more water arrives, so diluting it helps. The short list is where that flips. Some metals react with water and give off a gas that catches fire. Strong acid gives off so much heat when water hits it that the water boils and throws the acid back, which is why the rule is acid into water and never the other way round. Some solids give off a gas that is worse in the air than the drum was on the ground. None of this shows on the outside. The label is the only thing that says which list a drum is on, which is why a yard of drums with no labels is the emergency and not just a nuisance.

TAKEAWAY 13W

The usual answer is right for most things and dangerous for a few.

END OF THE DAY 11W

Provide a bounded plume assessment and decision triggers for evacuation zones.

What Dissolved in the River?

BEFORE THE DAY — FOLLOW

Adeyemi walks a bottle from the bank to the bench

A water sample is taken, cooled, carried and logged before anybody measures it. Adeyemi walks one the whole way. Stay with her, and watch how long the bottle spends warm in the van.

PLAN CARD 74W

Day three. Runoff reached the river overnight, and the town takes its drinking water out nine hundred metres downstream. Inês Ferreira has been out since four and wants it sampled properly. Today you decide where to sample, in what order, and whether to test the mud as well as the water. Test only the water and it comes back clean, and it honestly is clean today. The next flood stirs that mud back up.

BRIEFING 52W

Runoff from the accident enters the river. Some compounds remain in surface films, some dissolve, and others bind to sediment. The player must predict partitioning from molecular interactions and verify it with sampling. If the team samples only water, a persistent sediment reservoir may be missed until it re-enters the water supply.

WHAT YOU DECIDE 14W

Predict where each chemical class will be found and design a mass-balance sampling plan.

3.1 Where will it go?

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · PROTOCOL

SCENE 32W

Runoff from the fire has reached the river. Predict the wrong place and the team samples the wrong thing, and a clean water result gets read by the city as an all-clear.

GUIDE 55W

Four kinds of chemical, and four places in the river. Pair them by asking what water does with each one. Does it dissolve? Does it refuse to mix? Does it stick to the specks? A chemical that leaves the water has not left the river. Sample the wrong place and the city hears an all-clear.

ASK 10W

Match each kind of chemical to where it ends up.

BEHIND THE BUTTON 161W

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Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

some things dissolve in water and some refuse to mix with it

VERDICT 85W

Where a chemical goes follows from what it is like. Some dissolve straight into water and travel with it. Some refuse to mix at all, and sit on top as a film or sink underneath as a layer. And some stick hard to the specks of dirt floating in the water, ride those specks down, and end up in the mud. That last kind is the dangerous case for testing, because the water can come back clean while the river bed holds all of it.

TAKEAWAY 15W

Not finding something in the water does not mean it is not in the river.

3.2 Push it back out of the water

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · CHOICE

SCENE 29W

The filtered water holds 3.1 milligrams of dissolved metal in every litre. The limit is 0.5. The rig can put in eight times as much of the partner chemical.

GUIDE 54W

All four options give a number and a reason. The reasons are what differ. One says nothing changes. One says both go up together. Two say the metal falls, by different amounts. So work out how the two chemicals are tied to each other. The limit is well below where the water sits now.

ASK 11W

Eight times the partner chemical. What happens to the dissolved metal?

BEHIND THE BUTTON 169W

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TAKES AS READ

a dissolved metal and its partner chemical settle at a balance point together

VERDICT 84W

The metal and its partner are in a tug of war. When there is more of one, less of the other can stay dissolved, and the two move in opposite directions in step. Put in eight times as much partner and the dissolved metal falls to about an eighth. Three point one divided by eight is about 0.39, which is under the limit of half a milligram. A real rig would still have to check that the solid actually settles out and stays out.

TAKEAWAY 14W

Two things that settle together can be pushed apart by adding more of one.

3.3 Spend the river survey

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON ·
ALLOCATE

SCENE 32W

The survey has twenty bottles. The chemical barely dissolves, the river runs nine hundred metres from the spill to the water intake, and two samples taken so far disagree six times over.

GUIDE 49W

Twenty bottles is the whole survey. Each package you pick spends some of them. Watch the list of questions rather than the map: it shows which ones your plan can answer. Two samples already disagree six times over, so a plan that cannot explain why will be argued with.

ASK 11W

How do you spend twenty bottles without leaving a question unanswered?

BEHIND THE BUTTON 151W

Why the disagreement is the reason for the survey. Six times over between two samples means the chemical is not spread evenly through the water. It could be in a streak across the river, or deeper in some places, or changing with time. Each of those needs a different sampling pattern, and one bottle per place cannot tell them apart.

Why 'barely dissolves' matters. A chemical that barely dissolves travels partly as tiny droplets or stuck to grains of dirt, not just mixed into the water. So a filtered bottle and an unfiltered one answer different questions, and nine hundred metres is far enough for that to matter.

What the required questions are. Not everything you could ask. The ones a decision needs: is the intake affected now, will it be, and how much. A plan that answers other things well and misses one of those has spent the bottles badly.

TAKES AS READ

different places and different parts of a river answer different questions · a clean water sample cannot rule out a chemical sitting in the mud

VERDICT 78W

Three questions need answering and each needs its own bottles. What is reaching the intake now, which is what the city acts on today. What the river looked like before the spill, which is the only way to know whether a reading is unusual. And whether the chemical has left the water for the mud, which is where this kind goes. A pretty stained bank is interesting and answers none of the three. Twenty bottles cannot buy everything.

TAKEAWAY 20W

A sampling plan is the set of questions it can answer, and every bottle spent is a question given up.

3.4 Read the label, not the rumour

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM ·
CALLBACK · PROTOCOL

SCENE 36W

A fire at the freight yard has damaged several unlabelled drums, eighty metres from the river. Three different guesses are already on the radio. Firefighters are choosing what to spray based on whatever you tell them.

GUIDE 53W

The left side is four kinds of record. The right side is four things you could say from one. Ask of each record what it really pins down. Which elements? How many of each? What charge? A trade name pins down none of them. Firefighters are choosing what to spray from your answer.

ASK 12W

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TAKES AS READ

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VERDICT 78W

A formula says which elements are in something and how many of each. That is what decides how it behaves: what it dissolves in, what it reacts with, what it does to water. Change one of those numbers and it is a different substance with different dangers, even though the elements are the same. A trade name carries none of that. It is a word somebody chose, and some of the substances on this list react with water.

TAKEAWAY 20W

A name is a label. A formula says what something is made of, which is what decides how it behaves.

Predict where each chemical class will be found and design a mass-balance sampling plan.

The Concentration Problem

PLAN CARD 68W

Day six. The results are back as numbers, and the city is holding each one against a legal limit. Rie Nakamura wants every result run again before anything reopens. Marisol Reyes points out that four water systems are already shut on numbers nobody has questioned. Hospitals are on tankers. Today you turn readings into what was really in the river, and decide whether one result can be trusted.

BRIEFING 48W

City leaders ask whether the contaminant is "high," but samples differ in dilution, path length, and what else is in the sample. The team must turn signals into defensible concentrations. An incorrect concentration map can either close a safe water system or expose residents through a false reassurance.

WHAT YOU DECIDE 12W

Produce concentration maps whose units, detection limits, and quality controls are explicit.

4.1 Undo the dilution

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · BALLPARK

SCENE 42W

The lab watered the sample down before measuring it. So the number is what was in the vial, not the river. It is in mg/L, which means milligrams in every litre. The city is about to hold that number against a limit.

GUIDE 47W

Five numbers, and two belong to other questions. One is the smallest the instrument can see. One is the size of the bottle. Two are the volumes before and after watering down. Only how those two compare matters. The city will hold your answer against a limit.

ASK 7W

What was really in the river sample?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

watering a sample down lowers the reading by a known amount

VERDICT 74W

Samples get watered down so the reading lands inside the range the instrument can measure. That has to be undone before the number means anything about the river. Ten millilitres made up to a hundred is ten times weaker, so the river figure is ten times the reading. Report the reading as it stands and the river looks ten times cleaner than it is. Nothing about the number would show that anything was wrong.

TAKEAWAY 15W

What was done to a sample before it was measured is part of the answer.

4.2 Say what should happen, then check

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · VERIFY

SCENE 39W

The reading is above the range this meter was calibrated for. Calibration is the check against known samples. It is what turns a signal into a number. The meter still prints one, and it is going into a report.

GUIDE 56W

Lock the reading you expect after watering the sample down by half, before you do it. Then water it down and see what the meter says. Checking costs a verification slot, and spending it is the point. A meter prints a number for almost anything, and only the comparison shows whether this one can be trusted.

ASK 14W

What should the reading become after watering it down, and does the rerun agree?

BEHIND THE BUTTON 132W

Why watering it down is the test. If the reading really comes from this chemical, then half as much of it should give half the reading. That holds while the reading is inside the range the meter was calibrated for. Outside that range it does not halve, because the relationship there has stopped being straight.

What a reading outside the range is worth. The meter still prints one, with just as many digits and no warning. It is a guess dressed as a measurement, and a check like this one is the only way to catch it.

Why you lock the prediction first. Reading the watered-down sample and then deciding what you expected is not a test. Doing it in that order is what makes the second reading evidence instead of agreement.

TAKES AS READ

the reading goes up in step with how much is in the sample · watering something down by half halves what is in it

VERDICT 78W

Above the range it was set up for, an instrument still prints a number, and nobody knows what it means. Watering the sample down by half should halve the reading, and that is a prediction worth making before doing it. A reading of 1.34 should become about 0.67. The rerun gives 0.68, which is inside the trusted range and agrees with the prediction. Then the answer can be doubled back, and both halves of it have been checked.

TAKEAWAY 19W

Predict what a change should do, make it, and check. That is what turns a reading into a result.

4.3 Map the river with thirty samples

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · CHOICE

SCENE 25W

The city wants a picture of eleven kilometres of river. The laboratory can run thirty samples. The field kit is far rougher than the laboratory.

GUIDE 47W

Every option buys something real, and they trade against each other. Wide coverage costs detail. Repeats and checks cost samples. Thirty is all there is. Ask what decision each choice would improve. The city's decisions all happen in one place. Detail somewhere else is detail nobody uses.

ASK 10W

You cannot sample every metre. Where do the samples go?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a map is worth whatever the decisions made from it are worth

VERDICT 64W

There are two ways to waste this. Perfect chemistry at points nobody cares about, or lots of samples with a kit whose numbers nobody trusts. Both give a confident map that misleads. Every decision the city makes happens at the water intake, so that is where the detail belongs. Sampling where the water looks stained samples the staining, and this chemical has no colour.

TAKEAWAY 15W

Good numbers in the wrong places and rough numbers in the right ones both mislead.

4.4 From damaged drum to a first answer

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM ·
CALLBACK · SEQUENCE

SCENE 46W

In the mobile laboratory, the first sealed drum sample is on the bench. The markings on the outside are readable now, and opening the drum will scuff them off. Opening it also lets the air above the liquid escape. There is enough liquid for two tests.

GUIDE 50W

Every one of these can be done right now. So this is not about time. It is about what you can still do afterwards. Ask what each step uses up. The markings? The air above the liquid? The drops? The sample itself? There is only enough liquid for two tests.

ASK 13W

Put the four steps in order of what each one costs, cheapest first.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

some tests leave a sample as they found it, and some use it up

VERDICT 73W

The order is set by what each step costs. The markings on the outside survive only until somebody opens the drum, so they get photographed first, and it costs nothing. Sampling the air above the liquid takes what has already escaped, without opening anything. A test that gives the sample back can be repeated all week if it goes wrong. The test that uses the sample up answers best, and it answers once.

TAKEAWAY 14W

Do the tests that cost nothing before the ones that cannot be taken back.

END OF THE DAY 12W

Produce concentration maps whose units, detection limits, and quality controls are explicit.

The Heat Is Rising

BEFORE THE DAY — HUNT

Six sampling points, and four have markers

Every point on the river has to be found again next month or the numbers cannot be compared. Two markers are gone. A point nobody can find twice is a point that measures nothing.

PLAN CARD 71W

Day eleven. The fire has been out for eight days, and one storage bay is still getting warmer. Tomas Brandt has been reading it every hour since midnight. Fire Command wants the site released tonight, and the crews have been on shift for fourteen hours. Today you decide whether this is leftover heat or a reaction still running, and set up the watching either way. One reading cannot tell you which.

BRIEFING 42W

A storage zone continues warming after the visible fire is out. The team must distinguish stored heat from ongoing exothermic reaction and estimate whether cooling can keep up. If self-heating is mistaken for leftover warmth, responders may leave before runaway conditions develop.

WHAT YOU DECIDE 13W

Create an energy balance and choose a monitoring plan that detects self-heating early.

5.1 How much heat is stored in there

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · BALLPARK

SCENE 37W

The visible fire has been out for two hours, and the water bath beside the damaged bay is still warming. Fire Command wants the size of the heat before deciding whether the cooling on site is enough.

GUIDE 52W

Five numbers. One is for boiling the water away, which is not what happened. One is a temperature on another scale. Ask of each what it describes. How much water there is? What water is like? How much it changed? A small warming in a big tank is a lot of energy.

ASK 8W

How much energy has the bath taken in?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

different materials need different amounts of energy to warm by one degree

VERDICT 73W

Warming something by one degree takes energy, and how much depends on what it is and how much of it there is. Water is stubborn: every kilogram needs 4.2 units for each degree. Two thousand kilograms warming five degrees is forty-two thousand of them. Five degrees sounds like nothing, and against that much water it is an enormous amount of energy arriving from somewhere. That is what makes the small number worth calculating.

TAKEAWAY 14W

A small temperature rise across something heavy is a very large amount of energy.

5.2 Left-over heat, or something still going?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · DIAGNOSIS

SCENE 38W

Two bays saw the same fire. Ninety minutes after the cooling stopped, the east bay is cooling and the west bay is warming. The west bay is giving off more gas, and the hot spot has not moved.

GUIDE 54W

Six readings, and the east bay is the useful one. Same fire, nothing left running. Compare the two bays, reading by reading. A broken sensor would move both bays. Left-over heat can only drain away. Warming, with more gas, at one fixed spot is a different story. Fire Command releases the site on this.

ASK 9W

Is this heat left over, or something still running?

BEHIND THE BUTTON 156W

Why the quiet readings matter. It is easy to build a story around the number that is out of range. But most wrong explanations fail on something that did not move. If a cause would have pushed a second reading up and that reading is where it should be, the cause is not what happened, however well it explains the first one.

How to use the candidates. Take each one and ask what the panel would look like if it were true. Then compare that with the panel in front of you. Doing it in that order stops you fitting the story to the numbers, which is easy to do without noticing.

Why there is only one answer. More than one candidate will fit part of the panel, and that is on purpose. Only one fits every part of it. If two still look possible, find the reading they disagree about and let that reading choose.

TAKES AS READ

heat flows from hot to cold, and never the other way by itself

VERDICT 83W

Left-over heat can only drain away once the fire is out. That is what the east bay is doing, and it is the control. The west bay is doing the opposite: rising, with more gas coming off, and the hot spot sitting in one place at the drum stack. Nothing that is merely cooling behaves like that. Something in there is making heat. And a bad sensor would push both bays the same way, which is exactly why two bays are worth having.

TAKEAWAY 16W

A thing that keeps changing after its cause is removed has a cause of its own.

5.3 Control the self-heating

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · BALLPARK

SCENE 33W

Fire Command wants the site released tonight. Bay three has gone from thirty-one degrees to forty-four in eight hours, and it is climbing faster than it was. There are four tonnes in there.

GUIDE 48W

Four numbers. Two are temperatures the bay has read. One is the difference between them. One is the hours between. Ask which of those the question needs. A temperature is not a rise, and a rise is not a temperature. The site is released tonight on this number.

ASK 17W

If it keeps climbing at the same rate, what will bay three read in eight more hours?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate of change can be carried forward to guess what comes next

VERDICT 72W

Thirteen degrees in eight hours is the rate. Run it forward and you get another thirteen. That is about fifty-seven degrees by morning. The number turns an argument into a decision, because it can be held against the temperature where this material runs away. The catch is in the words "the same rate". It has been climbing faster each hour. So fifty-seven is the friendly answer, and the real one comes sooner.

TAKEAWAY 21W

A rising temperature is a rate, and a rate can be run forward to a time somebody has to act by.

5.4 What the bench is running now

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · SPOT

SCENE 23W

Samples queue on the bench all day. One priority says which are run first, and it is changed as the day goes on.

GUIDE 34W

Run what the priority asks for and leave the rest queued. The card on the bench changes during the day and nobody says so. What counts is the samples either side of a change.

ASK 12W

Run to the priority on the card, and keep checking the card.

BEHIND THE BUTTON 74W

Why the priority changes. Early it is whatever the commander is waiting for. Then it is whatever is about to go out of date, because a sample tested too late is not a result. Then it is drinking water, because people are asking.

Why it costs. The bench has one clean instrument. A sample run under the old priority uses it, and the one that mattered sits in the fridge going out of date.

TAKES AS READ

a laboratory works to a priority somebody can change

VERDICT 132W

Three priorities run across the day and each wants different samples. Whatever the commander is waiting for, then whatever is near its time limit, then anything from drinking water. A sample can match two at once, and that is what makes a change cost something. What the panel scores is the samples either side of a change. Most of the rack is wanted by neither priority, so leaving those queued proves nothing. The laboratory's own version is the twenty minutes after the schools ask about the taps. The priority has changed to drinking water, and the instrument is still working through river samples that were urgent an hour ago. And the cost is invisible later: a sample past its time limit comes out of the fridge looking just like a good one.

TAKEAWAY 18W

The cost of a rule that has changed is paid by whoever is still using the old one.

5.5 Float or sink

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM ·
CALLBACK · CHOICE

SCENE 33W

One drum is leaking into the bund. Nobody knows what the liquid is yet. Adaeze Okonjo needs to know whether it will float on the water already lying there, or sink under it.

GUIDE 52W

All four options are things you could measure about this liquid. Only one of them decides floating. Floating is about how heavy a litre is, next to a litre of water. Colour and smell tell you other things. So does the amount in the drum. The bund is filling while you decide.

ASK 9W

Which measurement decides whether the liquid floats on water?

BEHIND THE BUTTON 169W

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Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a litre of water weighs about a kilogram

VERDICT 101W

Floating is a comparison. A litre of water weighs about a kilogram, so a litre of anything lighter than that rides on top and a litre of anything heavier goes under. Nothing else on the list decides it. Colour and smell say nothing about weight. The amount in the drum does not change it either — a full drum and a spoonful of the same liquid float or sink the same way. The answer matters tonight, because a floating layer spreads across the bund and reaches the drain, while a sinking one sits at the bottom where the pump inlet is.

TAKEAWAY 20W

Whether something floats is a comparison, not a property: it is heavier or lighter than the same amount of water.

Create an energy balance and choose a monitoring plan that detects self-heating early.

The Water Changes pH

PLAN CARD 74W

Day fourteen. The water at the intake has turned acidic, and Reyes wants it neutralised this morning. Novák, who looks after the pipes, wants to know what a dose will do to them first. Today you work out how much acid is really there, and pick a dose that will not overshoot. There is one dosing pump, and no way to take a dose back out. Overshoot and metals come off the pipe walls.

BRIEFING 47W

The river intake becomes acidic, while one proposed treatment could overshoot into strongly basic conditions. The player must use acid-base equilibria and buffering to control pH without masking chemical risk. A pH correction can mobilize metals, damage infrastructure, or injure workers if the chemical context is ignored.

WHAT YOU DECIDE 13W

Select a controlled neutralization strategy and determine when buffer capacity will be exhausted.

6.1 How much more acidic

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · BALLPARK

SCENE 32W

The intake water reads four on the acid scale. Ordinary water reads seven. The plant is about to add something to bring it back, and wants the size of the problem first.

GUIDE 45W

Four numbers. Two are readings on the scale. One is the gap between them. The scale counts steps, and each step is ten times. So the gap is what the answer is built from, not the readings. The plant is sizing a dose from it.

ASK 12W

How many times more acidic is the intake water than ordinary water?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

each step down the acid scale means ten times more acid

VERDICT 73W

The acid scale is not like a ruler. Each step down means ten times more acid, not one more unit of it. So four is not a little below seven. It is three steps, and three steps is ten times ten times ten. A thousand times more acid than ordinary water. That is why a number that looks like a small change on paper needs a great deal more neutralising than anybody expects.

TAKEAWAY 15W

A scale where each step is ten times hides how big a difference really is.

6.2 Why the first dose overshoot

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · DIAGNOSIS

SCENE 43W

A dose worked out from the intake reading overshoot badly. The bench test is up beside a curve for a simple acid. That test is a titration: add a little at a time, until the water changes, and count how much it took.

GUIDE 54W

Six readings, and two are there to rule things out. The fresh meter checks, and the simple acid at the same reading. Compare the test curve against that simple acid. A long flat stretch is something a simple acid does not have. The first dose overshoot, so the answer has to explain that too.

ASK 13W

Which explanation accounts for the flat stretch and the dose it really took?

BEHIND THE BUTTON 156W

Why the quiet readings matter. It is easy to build a story around the number that is out of range. But most wrong explanations fail on something that did not move. If a cause would have pushed a second reading up and that reading is where it should be, the cause is not what happened, however well it explains the first one.

How to use the candidates. Take each one and ask what the panel would look like if it were true. Then compare that with the panel in front of you. Doing it in that order stops you fitting the story to the numbers, which is easy to do without noticing.

Why there is only one answer. More than one candidate will fit part of the panel, and that is on purpose. Only one fits every part of it. If two still look possible, find the reading they disagree about and let that reading choose.

TAKES AS READ

some acids hold most of their acid back, and let it out as you neutralise

VERDICT 72W

The reading measures the acid that is loose in the water at that moment. Some acids hold most of theirs back, and release more every time you neutralise what is loose. So the reading looks mild while the amount actually needed is five times bigger. That is what the long flat stretch on the test is: base going in and the reading barely moving, because more acid keeps arriving from the reserve.

TAKEAWAY 19W

A reading tells you what is loose right now. It does not tell you what is held in reserve.

6.3 Neutralise without overshooting

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · TRIGGER

SCENE 32W

The pump is primed and the operators are waiting on the valve. The water is acidic and the iron pipe is corroding. Every change to the pump shows up one reading later.

GUIDE 81W

One rule, written now and fixed once you release it. The base feed is pushing the tank up. Past pH 7.5 the water is bad again the other way. Here is the catch: a pump change shows on the meter one reading late, so the tank keeps rising after you stop the feed. Put your line below 7.5 by that one reading of rise. Do not put it so low that it fires on water which has not shown anything yet.

ASK 19W

The tank is climbing from 4.8 towards the 7.5 limit. At what pH do you stop the base feed?

BEHIND THE BUTTON 141W

What the delay does. Your correction lands on water that left the mixing point one interval ago. Push hard using an old reading and the correction arrives on top of one already in the pipe. That is what overshooting is, and the delay is the whole reason it happens.

Why both mistakes cost. The water is acidic and the pipe is iron, so dosing too little keeps eating the pipe. Dosing too much swings the tank the other way and the pump has to chase it back. Neither is free, and where you put the two thresholds is how you balance them.

Why the rules go first. A threshold changed while the readings arrive is not a rule, it is a description of what somebody already did. The operators at the valve need to know what will happen before it happens.

TAKES AS READ

the meter shows what the pump did one reading ago, not what it is doing now · each step on the acid scale is ten times, so equal steps are not equal amounts

VERDICT 80W

The pump acts now and the meter shows it later. So waiting until the meter reads what you want guarantees going past it, because more base is already on its way through the tank. Writing the rules first is what stops that. Slow the feed while the water can still absorb it, and stop it early enough that what is already in the pipe carries the reading the rest of the way. The threshold is a decision, not a description.

TAKEAWAY 11W

When the answer arrives late, the rule has to fire early.

6.4 How long before it reaches the school

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · CALLBACK · BALLPARK

SCENE 31W

A colourless cloud is drifting away from the yard toward two neighbourhoods. Police are holding the evacuation roads open. The wind is steady, and there is a school two kilometres downwind.

GUIDE 46W

Four numbers, and two of them are distances to other places. One is how wide the cloud is, which does not change when it arrives. Ask of each number what it measures. A distance? A speed? A size? Police are holding roads open on this answer.

ASK 8W

How long before the cloud reaches the school?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a cloud drifts at roughly the speed of the wind carrying it

VERDICT 76W

A cloud is carried by the wind, so it travels at roughly the wind's speed. Two thousand metres at four metres a second is five hundred seconds, which is a little over eight minutes. That is the whole decision. Eight minutes is enough to move a school if somebody starts now, and not enough if the first thing anybody does is argue. And if the wind picks up, the eight minutes shortens without anybody being told.

TAKEAWAY 16W

A hazard has a speed, and a speed turns a distance into the time you have.

END OF THE DAY 13W

Select a controlled neutralization strategy and determine when buffer capacity will be exhausted.

The Corrosion Failure

BEFORE THE DAY — CANVASS

Which streets are drinking the tap water

The exposure estimate assumes people stopped using the tap when they were told. Ask along the streets until enough answers have come back to say who stopped and who never got the notice.

PLAN CARD 74W

Day twenty-six. A section of the intake pipeline has failed, and Novák is standing in the hole in Third Street. That pipe carries the water a hundred and forty thousand people drink, and the city is on its reserves. Today you find where the metal is going, work out how fast, and choose the protection. Speed decides everything. The same loss over ten years is maintenance. Over a fortnight it is a burst main.

BRIEFING 36W

Acidic water and different metals threaten the intake pipeline. The player must identify redox couples and stop a small leak from becoming a system-wide failure. A hidden localized failure could interrupt drinking water during the emergency.

WHAT YOU DECIDE 14W

Identify the corrosion cell and choose controls that address both chemistry and electrical coupling.

7.1 Where the metal is going

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM ·
PROTOCOL

SCENE 34W

Acidic water is flowing through a section where two different metals are bolted together. An inspection has found one small break in the coating. The team wants the circuit understood before choosing a repair.

GUIDE 52W

The loop needs four things at once. A place where metal is lost. A place that takes the electrons. A path through the metal, and a path through the water. Each thing on the left gives or squeezes one of those. Ask which one. Break any single part and the loop stops.

ASK 13W

Match each thing happening at the metal to its part in the circuit.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

rusting is electricity, moving in a loop, with metal as one of the ingredients

VERDICT 75W

Rusting is not simply metal going bad. It is a circuit with four parts: a place losing metal, a place taking the electrons, a path through the metal, and a path through the water. Two different metals bolted together in water provide all four at once. And a coating with one small hole is worse than no coating, because everything that would have been spread over the whole pipe now goes through that one spot.

TAKEAWAY 13W

Rusting is a loop. Break any part of the loop and it stops.

7.2 How fast is the wall going

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · BALLPARK

SCENE 29W

A test piece from the intake has lost two kilograms of steel in a year, spread over four square metres. The wall that is left is six millimetres thick.

GUIDE 51W

Four numbers. One is the wall that is left, which belongs to the comparison. One is how much steel makes a millimetre over that patch. Ask which numbers the sum needs. And note what an average cannot see. The same loss in one small spot goes through the wall much sooner.

ASK 12W

How many years does it take to lose one millimetre of wall?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate only becomes a decision when it is compared with a thickness

VERDICT 80W

Two kilograms a year sounds alarming and means nothing on its own. Spread over four square metres of steel, one millimetre of thickness weighs about thirty-two kilograms, so two kilograms a year takes about sixteen years to eat through it. Six millimetres of wall is most of a century at that rate. That is the useful answer, and it is also a warning about averages. The same two kilograms arriving at one small hole goes straight through in a season.

TAKEAWAY 14W

An amount lost is not a warning until somebody turns it into a depth.

7.3 Protect the pipe

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 35W

The thinnest part of the intake pipe has three millimetres of wall left, and the loss has sped up eight times. The gallery stays cleared until the first control is chosen. Water is still flowing.

GUIDE 53W

Three of the four options act on a real part of the loop. One only moves paper. Ask which part each one removes. Then ask whether one small hole could beat it. The water is still flowing and three millimetres are left. The first fix has to stop the loop, not slow it.

ASK 7W

Which fix breaks the problem most directly?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a loop stops when any one part of it is broken

VERDICT 68W

The loop needs four parts, and breaking any one stops it. The water is acidic, which makes one path easy. The two metals bolted together supply the other. Separating those metals cuts the loop, and it can be done today. Changing the water works and is tangled up with everything else the plant has to do. A new coating helps, and it concentrates the attack wherever it fails.

TAKEAWAY 14W

Break the loop where you can break it today, and do the rest afterwards.

7.4 Neutral, while the works keeps feeding

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · HOLD

SCENE 34W

The tank has to sit in the middle of the pH scale before anything goes to the river. Acid keeps arriving from the works in batches, and the lime pump is the only control.

GUIDE 47W

Hold the tank at pH 7, inside the band on the meter. The band narrows as the discharge time comes closer. Each batch keeps arriving for as long as it is being pumped in, so set the lime to match the feed rather than chasing the meter.

ASK 12W

Hold the tank at pH 7 while the works keeps feeding it.

BEHIND THE BUTTON 66W

Why the middle is the target. Strong acid and strong base both do damage. Near the middle, water is safe for the river and for the people who work near it.

Why the last part is the hardest. The pH scale is not even. Near the middle, a small amount of lime moves the tank much further than the same amount did at the acid end.

TAKES AS READ

pH says how acidic or how basic something is

VERDICT 130W

Each batch of acid is not one push. It keeps arriving while it is being pumped, so the tank keeps falling until the lime is set to match it. That is why the pump is moved to a new setting and left there. The pH scale makes it harder than it sounds. It is not an even scale. Down at the acid end it takes a lot of lime to move the number one step, and near the middle the same lime moves it much further. So an operator who has learned how the tank feels at pH 4 will overshoot at 6.8 and end up adding acid back. The band narrows because what is measured is what leaves the works, not what the tank was doing an hour ago.

TAKEAWAY 12W

A feed you do not answer does not stop on its own.

7.5 What changes the cloud

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · CALLBACK · PROTOCOL

SCENE 25W

The weather station shows the temperature rising and the pressure falling, with the wind picking up. Crews report vapour pooling in a low channel downwind.

GUIDE 50W

Four changes in the field. Two of them change how much room the gas takes up. Two of them change where the gas goes. Ask which of the two each change does. Get that wrong and the map moves for the wrong reason. Vapour is pooling in a low channel.

ASK 10W

Match each change in the field to what it does.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a gas has a state, and it also has a place, and those change for different reasons

VERDICT 78W

Two different things are happening at once. Warming a gas makes it take up more room, and dropping the pressure around it does the same. That is about the gas itself. Whether it hugs the ground or lifts away depends on how heavy it is compared with air, which is a different question again. And where it goes is the wind's business entirely. Nothing about the gas changed when the wind turned. The place it was heading did.

TAKEAWAY 20W

Some things change what the gas is doing. Others change where it is going. They are not the same question.

END OF THE DAY 14W

Identify the corrosion cell and choose controls that address both chemistry and electrical coupling.

Remove It Without Making It Worse

BEFORE THE DAY — EVADE

The builder who wants a clean result today

He has been outside the office since eight wanting the site declared clear. Nothing is declared anywhere except on the form, and the second run is not back. Keep clear of him and get your round done.

PLAN CARD 75W

Day thirty-one. Kwabena Osei's pilot treatment works. Almost all of the compound is gone by the outlet, and he wants to scale it up this week. Nakamura points out that the test panel was written for that compound and nothing else. Today you work out what the treatment really did: what it removed, what it made, and where the rest went. A panel that is not looking for the new thing reports success either way.

BRIEFING 40W

Three treatment methods can lower the target contaminant, but one creates a toxic byproduct and another shifts the contaminant into sludge. The player must compare complete chemical consequences. A rapid treatment can create a less visible but more persistent hazard.

WHAT YOU DECIDE 14W

Choose a treatment train using contaminant removal, byproduct formation, waste fate, and operational reliability.

8.1 What did the treatment actually do?

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM · BALANCE

SCENE 36W

The pilot treated a batch holding a hundred kilograms of the chemical. The treated water, the sludge and the gas coming off have each been measured. The city is waiting to hear whether it was destroyed.

GUIDE 60W

Read every stream first, because reading is free. Then count only the masses that are the same chemical the batch went in with. A hundred kilograms went in, and the treated water, the sludge and the gas have each been measured. Whatever the counted streams do not account for is missing, and that gap is what the claim rests on.

ASK 14W

How much is missing from the measured streams, and does that support destroying it?

BEHIND THE BUTTON 147W

What counting a stream claims. Reading is neutral. Adding a stream to the total says the mass in it is the same chemical you started with. If a method reports a breakdown product as though it were the original, that claim is wrong and the total looks better than it is.

Why a gap is not proof of destruction. Mass that is missing from the measured streams went somewhere: out of a vent nobody measured, stuck to the inside of the vessel, or turned into something none of the three methods can see. Destroyed is one explanation, and the hardest one to check.

What the city is being told. Destroyed and moved somewhere else are very different for whoever ends up with the sludge or breathes the air. The ledger is what tells them apart, which is why the claim has to come out of the numbers.

TAKES AS READ

nothing simply disappears; it ends up somewhere · water getting cleaner does not say where the chemical went

VERDICT 77W

A hundred kilograms went in, and six are left in the water. That is not ninety-four destroyed. Seventy-one are in the sludge and three went off as gas, which is eighty accounted for. Twenty are missing, and the bench has already seen something new that nobody has measured. So the honest sentence is that the treatment moved most of it and changed some of it. Destroyed is a different claim, and this ledger does not support it.

TAKEAWAY 15W

Taking something out of one place is not destroying it. The ledger has to close.

8.2 Put the treatment in order

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · SEQUENCE

SCENE 30W

Four stages are on the table and the plant wants an order for them. Riverton has one intake, and it cannot run this experiment twice while the water is off.

GUIDE 50W

These four are not four separate machines. Each one changes what reaches the next. Ask what has to be there already. You cannot filter a solid before there is one. There is no solid until the metal is in the right form. Riverton has one intake and cannot try twice.

ASK 11W

Put the four stages in the order they have to run.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a filter can only take out something that is already solid

VERDICT 75W

Each stage hands something to the next. The first changes the metal into a form that can turn solid, because nothing later can make a solid out of something that will not make one. Changing the water then actually turns it solid. Only then is there anything for a filter to catch: filtering dissolved metal removes nothing at all. And the carbon goes last, because sending solids into a carbon bed clogs it within hours.

TAKEAWAY 14W

Each stage is ordered by what it leaves the next one to work with.

8.3 Choose what the pilot measures

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 35W

The pilot has room for one question, four weeks, and a tenth of the flow. All three methods claim to remove the chemical, and the full plant is committed in June whatever the pilot finds.

GUIDE 46W

All three methods remove the chemical. That is why they are all on the table. So measuring removal cannot separate them. Ask what each option would tell you that the others would not. Judged on removal alone, the winner is whichever makes the most leftovers fastest.

ASK 11W

All three remove it. What does the pilot have to measure?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

something that removes a chemical has to put it somewhere

VERDICT 72W

Every method here removes the chemical. That is why they are all on the table, and it is why measuring removal cannot choose between them. What separates them is what they leave behind: what the chemical turns into, and where the waste ends up. A pilot that measures only the target will pick whichever method makes the most new problem fastest, and it will look like a triumph while it does it.

TAKEAWAY 16W

Measure the thing that would tell the options apart, not the thing they all agree on.

8.4 Why the drum is worse in the afternoon

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · CALLBACK · CHOICE

SCENE 30W

The same open drum reads three times the vapour at two in the afternoon as it did at six in the morning. Nothing has been added and nothing taken out.

GUIDE 50W

All four options would explain a bigger reading. They differ in what changed. One changes the liquid. One changes the instrument. One changes how much is left. One changes only how much sits in the air. Nothing was added and nothing taken out. That is the fact that decides it.

ASK 9W

Same drum, same liquid, three times the vapour. Why?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a liquid is always sending some of itself into the air above it

VERDICT 78W

A liquid is always losing molecules into the air above it, and getting some back. How many are up there at once depends on temperature, and it climbs steeply. Ten or fifteen degrees can double it. So the drum that was safe to stand near at dawn is not the same drum at two in the afternoon, with nothing having been added or taken away. It is why measurements get taken at the hottest part of the day.

TAKEAWAY 13W

A measurement of a hazard belongs to the conditions it was taken in.

END OF THE DAY 14W

Choose a treatment train using contaminant removal, byproduct formation, waste fate, and operational reliability.

Can the Water Be Released?

PLAN CARD 69W

Day forty-five. The checking results are in, and one sits just under the limit. It is the re-run of a sample whose first quality check failed. Nakamura would like somebody to look at that. Reyes has schools and two hospitals on tankers, and wants the release signed this morning. Today you decide: release, release with conditions, or hold. Both mistakes are real, and they are not the same size.

BRIEFING 40W

Treatment results cluster near the decision threshold. The player must account for uncertainty, detection limits, and sampling representativeness before reopening the water system. A premature release exposes the city; an unnecessary hold can deprive hospitals and homes of essential water.

WHAT YOU DECIDE 14W

Make a transparent release decision with conditional monitoring and explicit treatment of borderline results.

9.1 Two candidates, one difference

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · CHOICE

SCENE 24W

Two substances fit the result equally well. Their formulas differ by one oxygen atom. An instrument that weighs molecules has already measured this sample.

GUIDE 44W

The two candidates differ by one oxygen atom. So ask which measurement notices one atom. Sort the options by what each one measures. How fast it travelled? How much there was? What it weighs? Two of these are real numbers about the wrong thing.

ASK 9W

Two candidates, differing by one oxygen. What settles it?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

what a molecule is made of decides what it weighs

VERDICT 75W

The two fit the first test equally well, so nothing about that test can choose. What they do differ in is weight: one oxygen atom is sixteen units, and there is an instrument here that weighs molecules directly. Sixteen units is an enormous difference to that instrument. The order things come out in follows chemistry rather than weight, and how big the peak is follows how much there is. Neither says which substance it is.

TAKEAWAY 14W

When two answers fit, use a measurement that depends on something they differ in.

9.2 Close the gaps before release

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · PROTOCOL

SCENE 29W

A number can be perfectly correct and still answer the wrong question. The pipes are about to be brought back up to pressure on the strength of this evidence.

GUIDE 49W

The left side is four separate holes in the evidence. Each has its own fix. Pair them by asking what was not tested. Sampling at the plant leaves the taps untested.

Running the tap first tests water nobody drinks. A machine that cannot see the limit cannot test it.

ASK 11W

Match each weakness in the evidence to what would close it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a sample tells you about the place and the moment it was taken

VERDICT 78W

Each of these is a different kind of gap and none of the fixes swaps for another. Sampling only where the water leaves the plant misses the taps people drink from. Sampling after running the tap for ages measures water nobody receives. An instrument that cannot see below the limit cannot test the limit. And one laboratory that both fixed the water and checked the water has marked its own homework, which no amount of good chemistry corrects.

TAKEAWAY 15W

A correct number is only useful if the sample it came from matches the decision.

9.3 Open, hold, or open with conditions

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · CLOUD

SCENE 38W

The pumps are idle while the board meets. The result is nine against a limit of ten, give or take two. Operators are waiting at the valves, and the city has already been given two different reopening times.

GUIDE 71W

The band on screen is every result the test allows, not just the best guess. It is nine against a limit of ten, give or take two, so part of the band is over the limit. Two pieces of follow-up work are ready. Each one does something different to the band, so watch its middle and its width separately. Apply what you would authorise, then say whether the rule is met.

ASK 25W

Can the whole band be brought under the limit, or only its middle? Move the bars to show the middle and one sigma either side.

BEHIND THE BUTTON 156W

What the band means. A number with a give-or-take on it stands for a range. Any value inside that range fits what was measured. The board is not asking whether the middle is under the limit. It is asking whether being over the limit has been ruled out.

Moving the middle is not the same as narrowing. A rerun that finds a mistake moves where the band sits. The band stays just as wide. Only new information makes it narrower: more samples, a better method, a second measurement. Both kinds of work are useful, and only one of them can clear a limit the width is crossing.

Why it matters here. The pumps are off and the city has already been told two different times. If the decision is made on the middle alone, the water goes back on with a real chance of being over the limit. The second apology costs more than the first.

TAKES AS READ

the band shown is the doubt that this decision has to live with · a middle below a limit is not a pass if the band crosses it

VERDICT 79W

Nine against a limit of ten looks like a pass until the band is drawn. Give or take two means the honest range runs up to eleven, and eleven is over. A rerun can move the middle without proving the band was too wide. Better sampling can narrow the band without moving the middle. Neither alone gets the top of the band under ten. Both together give 8.6 with a band of one, and a top edge of 9.6.

TAKEAWAY 15W

Moving the middle and narrowing the band are different moves, and a boundary needs both.

9.4 Water helps, or water makes it worse

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM ·
CALLBACK · BELT

SCENE 37W

The contractor lifts at four and the yard has to be sorted first. Some of these drums are made safer by water. Others react with water, and a hose is the worst thing that could reach them.

GUIDE 34W

Two bins. Ask what happens when water reaches it. Something that dissolves is made weaker by water. Something that reacts with water makes heat or gas, and then the hose has made things worse.

ASK 12W

Send each drum to the bin that says what water would do.

BEHIND THE BUTTON 66W

Why this is the first question. Water is what a fire crew has, so water is what usually arrives. For most of a yard that is right, and for a short list it is the worst possible move.

What reacting looks like. Heat, sometimes enough to boil the water. Gas coming off that was not there before. Both happen in seconds and neither can be undone.

TAKES AS READ

some things dissolve in water and some react with it

VERDICT 153W

Water is the usual answer to a spill because it is what everybody has. For most of a yard it is the right one. A salt or a weak acid gets weaker as more water arrives, so diluting it helps. The short list is where that flips. Some metals react with water and give off a gas that catches fire. Strong acid gives off so much heat when water hits it that the water boils and throws the acid back, which is why the rule is acid into water and never the other way round. Some solids give off a gas that is worse in the air than the drum was on the ground. None of this shows on the outside. The label is the only thing that says which list a drum is on, which is why a yard of drums with no labels is the emergency and not just a nuisance.

TAKEAWAY 13W

The usual answer is right for most things and dangerous for a few.

END OF THE DAY 14W

Make a transparent release decision with conditional monitoring and explicit treatment of borderline results.

Reopen the City

BEFORE THE DAY — TAG

Catch Okoye before the machine starts

Okoye has the list that says which bottles were blanks. Once the machine starts it runs for four hours. Without that list this morning's readings are numbers with no way to tell a real one from a clean one.

PLAN CARD 73W

Day fifty-two. The mayor wants one word tonight, and Sipho Mbeki has to put it on a screen behind him. Behind that word sits seven weeks of work, and not all of it is equally solid. Elena Stavros wants the long-term monitoring paid for before the emergency money closes on Friday. Nobody else has mentioned it. Today you take the claims one at a time, and decide what the city keeps paying for.

BRIEFING 44W

The final board must explain what was released, how chemicals moved and changed, which treatments worked, and what contamination remains in solids, infrastructure, and ecosystems. Without a complete fate and uncertainty record, the city cannot know whether the emergency is over or merely hidden.

WHAT YOU DECIDE 11W

Deliver a claim-by-claim chemical evidence package and a long-term monitoring plan.

10.1 What each claim has earned

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM ·
PROTOCOL

SCENE 33W

The city wants one sentence, and the chemistry does not support one. Turning all of it into a single green light is how a place gets reopened above a problem nobody wrote down.

GUIDE 50W

The evidence is in four different states at once. Each state needs a different sentence. Ask what would have to happen for each one to be settled. Another method? A way to measure it? More money? A trigger? Collapsing four states into one is how the sentence stops being true.

ASK 13W

Match each state of the evidence to what should be done with it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

evidence comes in strengths, and a claim is only as strong as the weakest step under it

VERDICT 67W

At the end of this, the evidence sits in four states, and each needs a different sentence. Something confirmed two separate ways can be written down as settled. Something likely, with no way to measure it yet, is provisional, and saying so keeps the record honest. Clean water above dirty mud is being managed rather than solved. And corrosion held inside a narrow window is a condition.

TAKEAWAY 11W

Readiness is claim by claim. There is no single green light.

10.2 Did the cell do what it should?

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · BALLPARK

SCENE 36W

The polishing cell ran for six hours. From the current it drew, it should have taken out about four and a half kilograms of metal. The water figures say it took out three and a half.

GUIDE 50W

Four numbers. Two are the current and the hours, and they were used to work out the most it could remove. That work is already done. Ask which two numbers the comparison needs. A share compares what happened with what was possible. Nothing above a hundred per cent is possible.

ASK 11W

What share of the possible removal did the cell actually manage?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the electricity a cell uses sets the most metal it could possibly remove

VERDICT 81W

The electricity going into a cell sets a ceiling. It cannot take out more metal than the current allows, however good it is. So the check is simple: what it actually removed, against what it could have. Three and a half out of four and a half is about eighty per cent, which is a real cell doing well. A figure over a hundred would mean the water measurements are wrong, or somebody has mixed up which tank they came from.

TAKEAWAY 14W

Comparing what happened with what should have happened is how a claim gets checked.

10.3 What the city keeps

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 31W

The temporary laboratory is being packed up as the emergency money ends. The plant is running. The mud still holds about seventy-one kilograms. One monitoring station has funding after next month.

GUIDE 50W

All four options are fair uses of the last money. They protect different things. Ask what nobody would notice if it were not funded. Seventy-one kilograms are still in the mud. None of it was destroyed. It went quiet. What lasts after a spill is the part that went quiet.

ASK 13W

The emergency money is about to move elsewhere. What does the city keep?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

falling below a limit is not the same as being gone

VERDICT 76W

Nothing here was destroyed. Seventy-one kilograms are in the mud, twenty were never accounted for, and something new is still being worked out. All of it is below a limit rather than gone. Watching the water, the mud and the air is the only thing on this list that would notice any of it coming back. Corrosion control protects the pipes, which is a different problem. Keeping the records costs almost nothing and should happen anyway.

TAKEAWAY 15W

What outlasts an emergency is the part that went quiet, not the part that stopped.

10.4 Where will it go?

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · CALLBACK · PROTOCOL

SCENE 32W

Runoff from the fire has reached the river. Predict the wrong place and the team samples the wrong thing, and a clean water result gets read by the city as an all-clear.

GUIDE 55W

Four kinds of chemical, and four places in the river. Pair them by asking what water does with each one. Does it dissolve? Does it refuse to mix? Does it stick to the specks? A chemical that leaves the water has not left the river. Sample the wrong place and the city hears an all-clear.

ASK 10W

Match each kind of chemical to where it ends up.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

some things dissolve in water and some refuse to mix with it

VERDICT 85W

Where a chemical goes follows from what it is like. Some dissolve straight into water and travel with it. Some refuse to mix at all, and sit on top as a film or sink underneath as a layer. And some stick hard to the specks of dirt floating in the water, ride those specks down, and end up in the mud. That last kind is the dangerous case for testing, because the water can come back clean while the river bed holds all of it.

TAKEAWAY 15W

Not finding something in the water does not mean it is not in the river.

END OF THE DAY 11W

Deliver a claim-by-claim chemical evidence package and a long-term monitoring plan.

THE CLOSING CARD

The plant came back online on day sixty-one, and the city has been drinking its own water since. Every claim in the package held: the compound identified by two methods, the plume mapped, the sediment reservoir bounded, and ten years of monitoring funded before the emergency budget closed.

The fence came down on a Thursday. Both neighbourhoods were home by the weekend, and the river survey still samples the same four stations every month.

Card text only — no options, boards or answers. 7,995 words of card prose across 41 stops. Generated from themes/contamcity_ms by tools/make-cardbook.mjs.